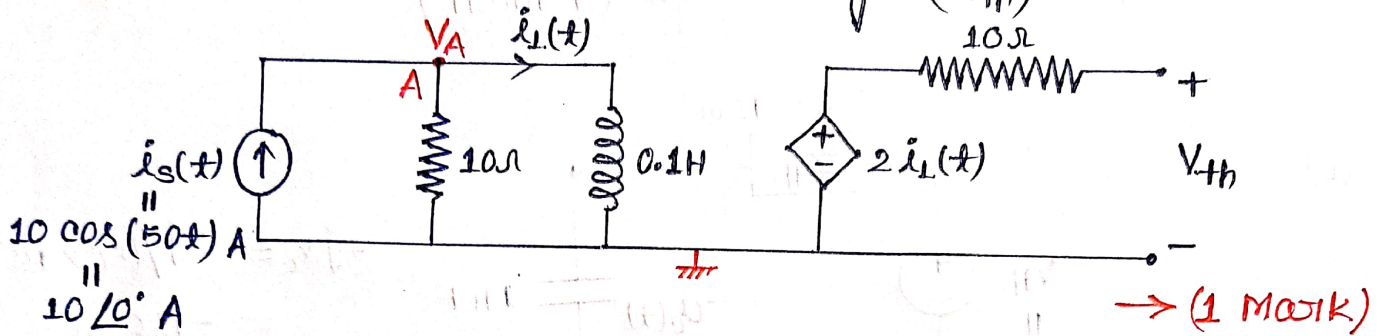


CTD MID-SEM EXAM
SOLUTION (SET-A)

Sol(1)÷

(a) Thevenin equivalent across the terminal of the capacitor

Step(I)÷ Calculation of Thevenin voltage (V_{th})



Apply nodal analysis at node-A,

$$\frac{V_A}{10} + \frac{V_A}{jX_L} = 10 \angle 0^\circ$$

$$V_A \left(\frac{1}{10} + \frac{1}{j \times 50 \times 0.1} \right) = 10 \angle 0^\circ$$

$$V_A \left(\frac{2+j}{10j} \right) = 10 \angle 0^\circ$$

$$V_A = 10 \angle 0^\circ \cdot \left(\frac{10j}{2+j} \right) = 10 \angle 0^\circ \times 4.472 \angle 63.43^\circ$$

$$V_A = 44.72 \angle 63.43^\circ \text{ V} = 44.72 \cos(50t + 63.43^\circ) \text{ V}$$

→ (2 Mark)

Now, $V_{th} = 2i_L(t)$

$$= 2 \left\{ \frac{V_A}{j \times 50 \times 0.1} \right\} = 2 \left\{ \frac{44.72 \angle 63.43^\circ}{j5} \right\}$$

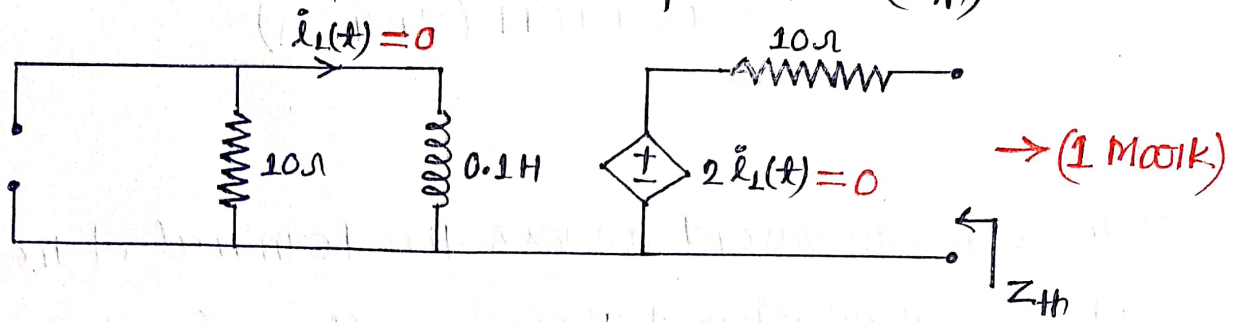
$$= 2 \left\{ \frac{44.72 \angle 63.43^\circ}{5 \angle 90^\circ} \right\}$$

$$= 17.89 \angle -26.57^\circ \text{ V}$$

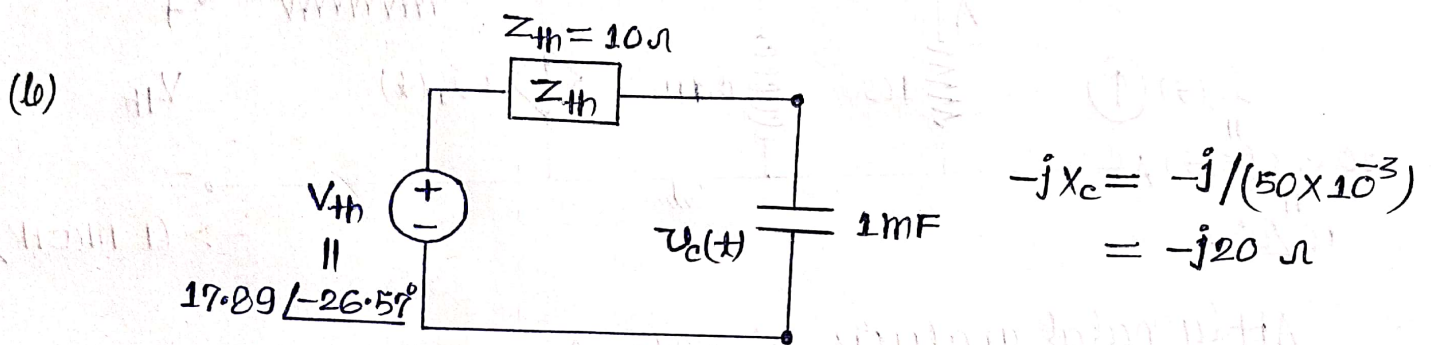
$$= 17.89 \cos(50t - 26.57^\circ) \text{ V}$$

→ (2 Mark)

Step(II) ÷ Calculation of Thevenin impedance (Z_{th})



∴ $Z_{th} = 10\Omega$ → (1 Mark)



$$v_c(t) = \left(\frac{-jX_c}{Z_{th} - jX_c} \right) V_{th}$$

$$= \left(\frac{-j20}{10 - j20} \right) 17.89 \angle -26.57^\circ$$

$$= 0.89 \angle -26.57^\circ \cdot 17.89 \angle -26.57^\circ$$

$$= 15.92 \angle -53.14^\circ \text{ V}$$

$$= 15.92 \cos(50t - 53.14^\circ) \text{ V} \rightarrow (2 \text{ Mark})$$

$$i_L(t) = \frac{V_{th}}{2}$$

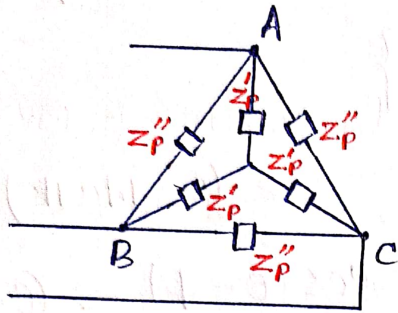
$$= \frac{17.89 \angle -26.57^\circ}{2}$$

$$= 8.95 \angle -26.57^\circ \text{ A}$$

$$= 8.95 \cos(50t - 26.57^\circ) \text{ A} \rightarrow (1 \text{ Mark})$$

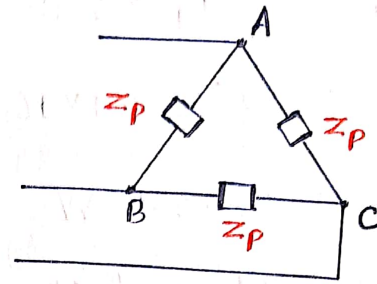
Sol(2)-e

Connection of loads-



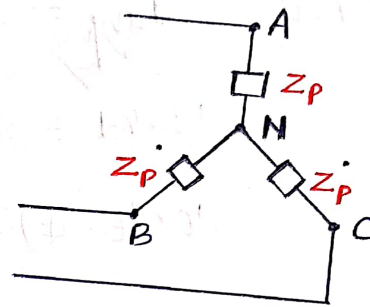
$$Z_p' = (5 + j3) \text{ k}\Omega$$

$$Z_p'' = (12 + j) \text{ k}\Omega$$



$$Z_p = (7.05 + j1.94) \text{ k}\Omega$$

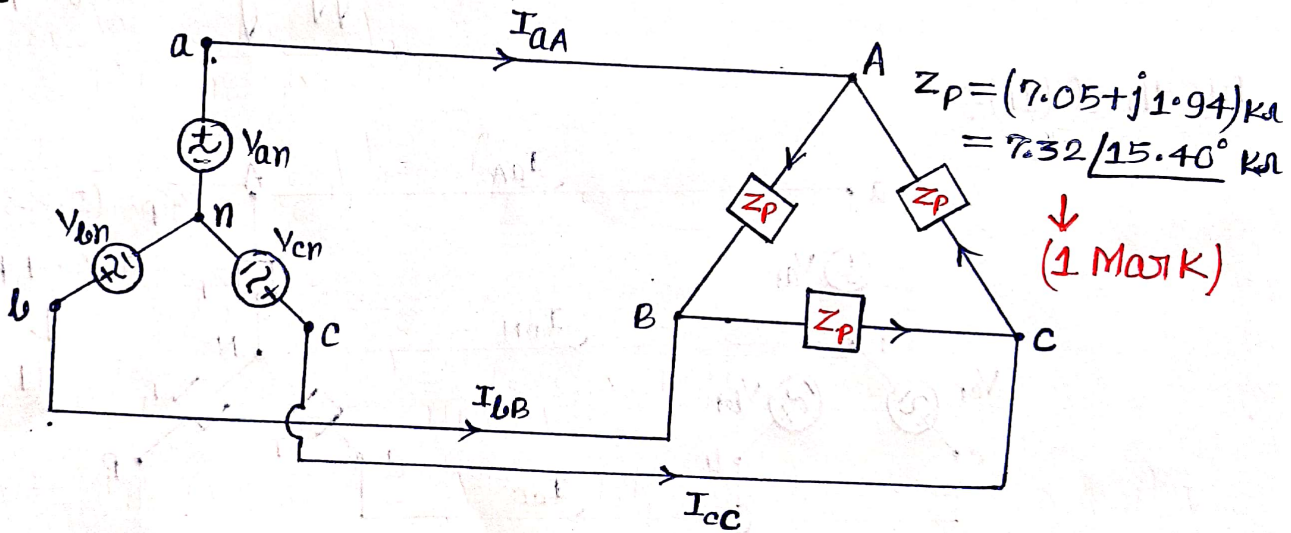
$$= 7.32 / 15.40^\circ \text{ k}\Omega$$



$$Z_p = (2.35 + j0.65) \text{ k}\Omega$$

$$= 2.44 / 15.46^\circ \text{ k}\Omega$$

Method (I)-



(a)

Line Voltage, $V_L = 240 \text{ V rms}$ (Given)

Phase Current, $I_p = \frac{V_L}{|Z_p|} \rightarrow (0.5 \text{ Mark})$

$$= \frac{240}{7.32} \times 10^{-3} = 32.79 \text{ mA rms} \rightarrow (1 \text{ Mark})$$

Line Current, $I_L = \sqrt{3} \cdot I_p \rightarrow (0.5 \text{ Mark})$

$$= \sqrt{3} \times 32.79 = 56.79 \text{ mA rms} \rightarrow (1 \text{ Mark})$$

(b) Average power taken by the combined load,

$$P_{avg} = 3(I_p^2 \times 7.05 \times 10^3) \rightarrow (1 \text{ Mark})$$

$$= 3 \left\{ (32.79 \times 10^{-3})^2 \times 7.05 \times 10^3 \right\}$$

$$= 22.74 \text{ W}$$

$\rightarrow (2 \text{ Mark})$

(c) Power factor of the load = PF = $\cos(\theta - \phi) \rightarrow (0.5 \text{ Mark})$

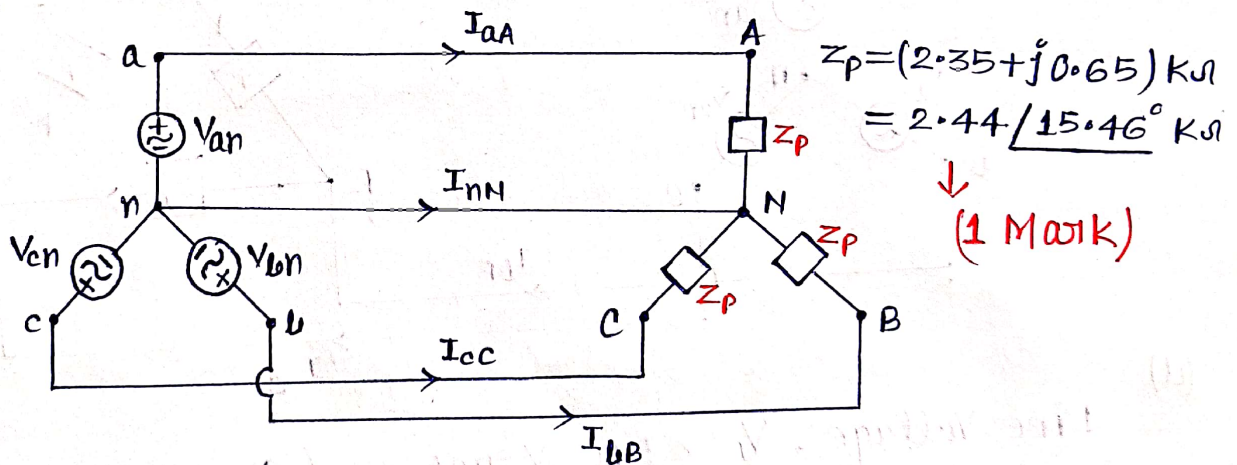
As we know - $P_{avg} = \sqrt{3} V_L I_L \cos(\theta - \phi) \rightarrow (0.5 \text{ Mark})$

$$\therefore 22.74 = \sqrt{3} \times 240 \times 56.79 \times 10^{-3} \times \cos(\theta - \phi)$$

$$\cos(\theta - \phi) = \frac{22.74}{\sqrt{3} \times 240 \times 56.79 \times 10^{-3}} = 0.96$$

$\therefore \text{PF} = \cos(\theta - \phi) = 0.96 \text{ lagging} \rightarrow (2 \text{ Mark})$

Method (II) ÷



(a) Line Voltage, $V_L = 240 \text{ V rms}$ (Given)

Phase Voltage, $V_p = \frac{V_L}{\sqrt{3}} \rightarrow (0.5 \text{ Mark})$

$$= \frac{240}{\sqrt{3}} = 138.56 \text{ V rms} \rightarrow (1 \text{ Mark})$$

Line Current, $I_L = \frac{V_p}{|Z_p|} \rightarrow (0.5 \text{ Mark})$

$$= \frac{138.56}{2.44} \times 10^{-3} = 56.79 \text{ mA rms}$$

(b) Average power taken by the combined load,

$$P_{avg} = 3(I_L^2 \times 2.35 \times 10^3) \rightarrow (1 \text{ Mark})$$

$$= 3 \left\{ (56.79 \times 10^{-3})^2 \times 2.35 \times 10^3 \right\}$$

$$= 22.74 \text{ W} \rightarrow (2 \text{ Mark})$$

(c) As we know - $P_{avg} = \sqrt{3} V_L I_L \cos(\theta - \phi) \rightarrow (0.5 + 0.5 \text{ Mark})$

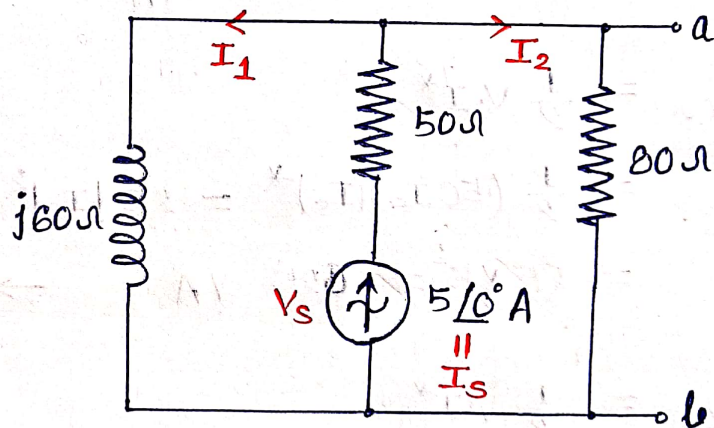
$$22.74 = \sqrt{3} \times 240 \times 56.79 \times 10^{-3} \times \cos(\theta - \phi)$$

$$\cos(\theta - \phi) = 0.96$$

$\therefore \text{PF} = \cos(\theta - \phi) = 0.96 \text{ lagging} \rightarrow (2 \text{ Mark})$

Sol(3):

$$\omega = 100 \text{ rad/s (Given)}$$



Apply KVL,

$$I_1 + I_2 = 5\angle 0^\circ \quad \text{--- (1)}$$

$$j60 I_1 = 80 I_2 \quad \text{--- (2)}$$

By solving eqⁿ(1) & eqⁿ(2), we get -

$$I_1 = 4\angle -36.87^\circ \text{ A}$$

$$I_2 = 3\angle 53.13^\circ \text{ A}$$

} $\rightarrow (2 \times 0.75 \text{ Mark})$

$$V_s = 50 I_s + 80 I_2$$

$$= 50 \times 5 \angle 0^\circ + 80 \times 3 \angle 53.13^\circ = 438.29 \angle 25.98^\circ \text{ V}$$

→ (0.5 Mark)

(a) Hence, $V_s = 438.29 \angle 25.98^\circ \text{ V}$

$$I_s = 5 \angle 0^\circ \text{ A}$$

$$\therefore \text{PF} = \cos(\theta - \phi) \rightarrow (0.25 \text{ Mark})$$

$$= \cos(25.98 - 0) = 0.90 \text{ lagging} \rightarrow (0.75 \text{ Mark})$$

(b) $|S_{j60\Omega}| = \left| \frac{1}{2} V \cdot I^* \right|$

$$= \left| \frac{1}{2} (j60 \times I_1) (I_1)^* \right| = |j30 \times |I_1|^2|$$

$$= |j30 \times 4^2| = 480 \text{ VA}$$

→ (0.25 + 0.75 Mark)

$$|S_{50\Omega}| = \left| \frac{1}{2} V \cdot I^* \right|$$

$$= \left| \frac{1}{2} (50 I_s) (I_s)^* \right| = 25 |I_s|^2$$

$$= 25 \times 5^2 = 625 \text{ VA}$$

→ (0.25 + 0.75 Mark)

$$|S_{80\Omega}| = \left| \frac{1}{2} V \cdot I^* \right|$$

$$= \left| \frac{1}{2} (80 I_2) (I_2)^* \right| = 40 |I_2|^2$$

$$= 40 \times 3^2 = 360 \text{ VA}$$

→ (0.25 + 0.75 Mark)

(c) Average Power supplied by the source,

$$(P_{\text{avg}})_{\text{source}} = \frac{1}{2} V_s I_s \cos(\theta - \phi)$$

$$= \frac{1}{2} \times 438.29 \times 5 \times 0.90$$

$$= 986.15 \text{ W}$$

→ (0.25 + 0.75 Mark)



Average power delivered to each passive elements,

$$(P_{avg})_{j60\Omega} = \frac{1}{2} |I_1|^2 \cdot 0 = 0 \text{ W}$$

$$\begin{aligned}(P_{avg})_{50\Omega} &= \frac{1}{2} |I_5|^2 \cdot 50 \\ &= \frac{1}{2} \times 5^2 \times 50 = 625 \text{ W}\end{aligned}$$

$$\begin{aligned}(P_{avg})_{80\Omega} &= \frac{1}{2} |I_2|^2 \cdot 80 \\ &= \frac{1}{2} \times 3^2 \times 80 = 360 \text{ W}\end{aligned}$$

→ 3 × (0.25 + 0.75 Mark)

CTD MID-SEM EXAM
SOLUTION (SET-B)

NOTE: Refer to Set-A solution for calculation, figure, marks distribution

Sol(1): $I_1 = 4 \angle -36.87^\circ \text{ A}$

$$I_2 = 3 \angle 53.13^\circ \text{ A}$$

} $\rightarrow (2 \times 0.75 \text{ Mark})$

$$V_S = 438.29 \angle 25.90^\circ \text{ V} \rightarrow (0.50 \text{ Mark})$$

(a) $(P_{\text{avg}})_{\text{source}} = 906.15 \text{ W} \rightarrow (0.25 + 0.75 \text{ Mark})$

(b) $(P_{\text{avg}})_{j60\Omega} = 0 \text{ W}$

$$(P_{\text{avg}})_{50\Omega} = 625 \text{ W}$$

$$(P_{\text{avg}})_{80\Omega} = 360 \text{ W}$$

} $\rightarrow 3 (0.25 + 0.75 \text{ Mark})$

(c) $|S_{j60\Omega}| = 400 \text{ VA}$

$$|S_{50\Omega}| = 625 \text{ VA}$$

$$|S_{80\Omega}| = 360 \text{ VA}$$

} $\rightarrow 3 (0.25 + 0.75 \text{ Mark})$

(d) $\text{PF} = 0.90 \text{ lagging} \rightarrow (0.25 + 0.75 \text{ Mark})$

Sol(2): Refer to Set-A solution

Sol(3): Refer to Set-A solution